Reference (setting)	Primary aim of study	Sex, strain and species (n/group)	Age and/or bodyweight	Depression model	Pharmacological intervention (dose, route of administration)	Main findings
Alboni et al., 2018 (Italy)	Examination of NSAIDs ability to increase effectiveness of antidepressants	Male S-D rats (5-10)	200-250g	Minor unavoidable stress	FLX (5mg/kg, i.p) + Flurbiprofen (5mg/kg, i.g) FLX (5mg/kg, i.p)+ Celecoxib (5mg/kg, i.g) FLX (5mg/kg, i.p)+ ASA (11.25-45mg/kg, i.p) 7 days of treatment	FLX + ASA → ↓escape deficit.Amplitude of antidepressant effect of ASA is dose dependant FLX + celecoxib → (↓) dep FLX + flurbiprofen → Ø
Ballok et al., 2006 (Canada)	Examination of ibuprofen's effects on lupus-induced brain atrophy and behavioural dysfunction	Male MRL-lpr mice (10)	5-19 weeks	MRL-lpr lupus model	Ibuprofen (375 ppm in food chow) 14 weeks	Ibuprofen → no effect on lupus induced despair or on lupus induced microgliosis
Borges et al., 2014 (Spain)	To examine whether NSAIDs can alleviate arthritic pain induced anxiety and	Male S-D rats (5-6)	250-300g	MoA induced by CFA injection	Diclofenac (10mg topically) twice daily for 3-5 days	Diclofenac → ↓MoA induced despair and restored ERK1/2 activation levels in the brain
Brunello et al., 2006 (Italy)	Evaluate the antidepressant effect of coadministation with ASA and FLX	Male S-D rats (7-8)	150-175 g	Chronic escape deficit model	FLX (5mg/kg, i.p) ASA (22.5 or 45mg/kg, i.p) testing after 1, 2 and 3 weeks of treatment	FLX + ASA → ↓escape deficit FLX → ↓escape deficit ASA → Ø Faster therapeutic effect of FLX + ASA compared to FLX alone
De La Garza II 2005 (USA)	To examine the effect of diclofenac on LPS-induced anhedonia	Male Wistar rats (8-9)	8 weeks, 225-250g	LPS	Diclofenac (2.5mg/kg, s.c) 2.5 h before testing LPS (20ug/kg, i.p) 2h before testing	Diclofenac → ↓LPS-induced anhedonia but had no effect on LPS-induced elevation of plasma IL-1β

de Munter et al., 2020 (Russia/Netherlands)	To examine abnormal behaviour in a genetic model of frontotemporal lobar degeneration (FUS[1-359]-tg mice)	Male FUS[1-359]-tg mice (9-15)	9-10 weeks	FUS[1-359]-tg mice	Celecoxib (30mg/kg/day, p.o) 3 weeks treatment	Celecoxib in WT → Ø despair Celecoxib in FUS[1-359]-tg → ↓despair and normalized IL-1B expression and microgliosis in the brain
Deak et al., 2005 (USA)	To examine the effects of anti-inflammatory agents and LPS on FST	male S-D rats (8)	350-400g	FST	Indomethacin (1 or 10mg/kg, i.p), 60 min before FST	Indomethacin → Ø
Feng et al., 2020 (China)	To examine effects of MA and celecoxib on depressive-like behaviour and microgliosis	Male C57BL/6 mice (8)	7 weeks, 20-25g	28 day CMS	MA (5mg/kg/day, i.p) celecoxib (10mg/kg/day, i.p) 28 days treatment	MA →↓CMS induced anhedonia, despair and microgliosis Celecoxib→↓CMS induced anhedonia, despair and microgliosis. MA and celecoxib inhibit ERK1/2 and P38 MAPK activation and iNOS expression.
Fischer et al., 2015 (Denmark)	To investigate of IFN-α on depression and TRP metabolites	Male S-D rats (10)	300g	IFN-α (60000 IU/kg/day), 7 days	Celecoxib (16mg/kg/day, p.o), 7 days treatment	Celecoxib → (↓)INF-α induced despair and altered KYN/TRP ratio
Guan et al., 2014 (China)	To investigate antidepressant effects of ASA	Male S-D rats (7)	220-250g	FST	ASA (6, 12, 25 and 50mg/kg, i.p) treatment 24, 5 and 1 h before FST	ASA (25 and 50mg/kg) → ↓FST-induced depression and normalised peripheral TNF-α and IL-6 levels but had no effect on FST-induced CORT
Guevara et al., 2015 (Venezuela)	To evaluate the association between spinal PGE2 and thermal hyperalgesia following stress.	Male S-D rats (?)	150-300g	Repeated swim stress	Ketoprofen (6mg/kg, i.p) Meloxicam (1mg/kg, i.p) treatment 30 min before swim session (3 sessions in total)	Ketoprofen → ↓repeated swim test induced despair Meloxicam → ↓repeated swim test induced despair. Ketoprofen and meloxicam reduced serum CORT and normalised inflammation induced spinal
Guo et al., 2009 (China)	To examine the antidepressant effects of celexocib	Male S-D rats (?)	200-220g	CMS	Celecoxib (2, 8 or 16mg/kg/day, p.o) 21 days Celecoxib (16mg/kg, p.o.) single dose 1 h before stress exposure	Celecoxib (8 and 16 mg/kg/day) → ↓CMS induced anhedonia and brain COX-2 and PGE2 elevation

Hu et al., 2010 (China)	To examine whether neuropathic pain can induce cognitive dysfunction	Male S-D rats (10)	160-180g	L5 spinal nerve transection	Lornoxicam (2 mg/kg/day, i.p) on postoperative days 7-28	Lornoxicam → no effect on neuropathic pain induced despair
Kurhe et al., 2014 (India)	To examine if celexocib has antidepressant effects on obese mice	Male Swiss albino mice (6)	20-25g	HFD	Celexocib (10mg/kg/day, p.o) 28 days	Celecoxib → ↓HFD induced despair and anhedonia
Luo et al., 2017 (China)	To examine the role of COX-2 in learning and memory	? S-D rats (10)	8 weeks, 180-200g	CMS	Meloxicam (3mg/kg/day, p.o.) 21 days	Meloxicam → ↓CMS-induced anhedonia
Luo et al., 2020 (China)	To screen target proteins regulated by nimesulide in hippocampus of CUMS rats	Male S-D rats (20)	180-220g	CMS	Nimesulide (12mg/kg/day, p.o) 21 days	Nimesulide → ↓CMS-induced despair and neuroinflammation
Maciel et al., 2013 (Brazil)	To characterize depression-like behaviour in an inflammation model induced by CFA	Male Swiss mice (6-8)	25-30g	CFA	Celecoxib (3, 15 or 30 mg/kg/day p.o) 7 days	Celecoxib (15 and 30mg/kg) → ↓CFA-induced despair and neuroinflammation
Martin-de-Saavedra et al., 2013 (Spain)	To examine the role of Nrf2 in depression	Male C57Bl/6 mice Nrf2 WT and KO (10)	3-4 months	Nrf2 knockout	Rofecoxib (2mg/kg/day i.p) 7 days	Rofecoxib → ↓Nrf2 KO-induced despair

Mesripour et al., 2019 (Iran)	To examine the effects of NSAIDs on IFN-α induced depression	Male albino mice (6)	24-28g	IFN-α (16 x 10^5 IU/kg, s.c) 6 days	Celecoxib (25 or 50 mg/kg/day i.p) Ibuprofen (50 or 75 mg/kg/day i.p) Indomethacin (12.5 or 25 mg/kg/day i.p) 1 or 6 days of treatment	Celecoxib (50 but not 25mg/kg) → antidep effect alone in FST Celecoxib (25 and 50mg/kg) → ↓IFN-α induced despair Celecoxib (50 acute not 25 subacute) → ↓IFN-α induced anhedonia Ibuprofen (50 but not 75mg/kg) → antidep effect alone in FST Ibuprofen (50 acute and subacute) → ↓IFN-α induced despair and anhedonia Indomethacin (12.5 and 25 mg/kg) → antidep effect alone in FST Indomethacin (25mg/kg acute) → ↓IFN-α induced despair and anhedonia
Morgese et al., 2018 (Italy)	To examine the effects of celecoxib on Aβ induced	Male Wistar rats (8)	275-300g	ICV Aβ	Celecoxib (15mg/kg/day s.c.) 8 days	Celecoxib → ↓Aβ-induced despair and normalise 5HT levels
Nemeth et al., 2014 (USA)	To examine the effect of meloxicam on HIV-induced depression	Female WT and HIV-1 transgenic rats (5-9)	48-58 days	HIV-1	Meloxicam (1mg/kg/day, p.o) 33 days	Meloxicam → no effect on HIV-induced despair but prevents neuroinflammation
Nemeth et al., 2016 (USA)	To examine inflammatory mechanism in microembolism-induced depressive-like behaviours	Male Wistar rats (12-13)	3 months	Microembolism	Meloxicam (1mg/kg/day i.p) 14 days	Meloxicam → ↓microembolism-induced anhedonia and normalise expression of Spp1
Norden et al., 2015 (USA)	To examine the effect of ibuprofen on cancer induced behavioural alterations	Female BALB/c x DBA/2 F1 (CD2F1) mice (?)	10 weeks 20-22g	Cancer tumor	Ibuprofen (10mg/kg/day p.o) 10 days	Ibuprofen → ↓tumor-induced despair and peripheral + central inflammation Ibuprofen alone → no effect on despair

Pavlock et al., 2021 (USA)	To examine neuroinflammation in OVX mice treated with the chemotherapy drug CP and	Female C57Bl/6 mice (6-8)	9-10 weeks, 16-20g	CP	Naproxen (375 ppm in diet) 3 weeks	Naproxen → no effect on despair neither alone nor in OVX/CP animals
Perveen et al., 2018 (Pakistan)	To examine the role of COX-2 inhibitors on stress-induced behavioural alterations	Male S-D rats (6)	150-200g	7 days mild stress	Indomethacin (7.5mg/kg/day, i.p) Diclofenac (5 mg/kg/day, i.p) 7 days treatment	Indomethacin → ↓despair alone and after stress and normalised plasma CORT levels Diclofenac → ↓despair alone and after stress and normalised plasma CORT levels
Qadeer et al., 2018 (Pakistan)	To examine the role of ibuprofen and lavender oil on stress induced behavioural abnormalities	? Wistar rats (6)	150-200	Restraint stress	Ibuprofen (2mg/kg/day p.o) 7 days of treatment	Ibuprofen → ↓stress-induced despair. No effect in unstressed animals
Saleh et al., 2014 (Egypt)	To examine the effect of ibuprofen in BCG induced depressive-like behaviour	Male Swiss albino mice (7-10)	22-30g	BCG	Ibuprofen (40mg/kg/day p.o) 2 weeks treatment	Ibuprofen → ↓BCG induced despair in both TST and FST
Salmani et al., 2020 (Iran)	To examine the effects of ibuprofen on LPS induced brain alterations	Male BALB/c mice (10-11)	2 months, 25-35g	LPS (250ug/kg/day, 6 days)	Ibuprofen (40mg/kg/day i.p) 10 days treatment	Ibuprofen+vehicle → ↓despair Ibuprofen+LPS → Ø despair but decreased neuroinflammation
Santiago et al., 2014 (Brazil)	To examine the antidepressant-like effects of celecoxib and piroxicam	Male Wistar rats (8-11)	3 months, 280-320g	3 weeks CMS	Celecoxib (10mg/kg p.o) Piroxicam (10mg/kg p.o) Treatment 1, 5 and 24h before testing or daily for 21 days	Celecoxib alone → ↓despair and altered monoamine expression Piroxicam alone → ↓despair and altered monoamine expression Celecoxib → ↓CMS-induced anhedonia Pirocixam → ↓CMS-induced anhedonia
Santiago et al., 2015 (Brazil)	To investigate the antidepressant effects of piroxicam on 6-OHDA lessioned rats	Male Wistar rats (6-9)	280-320g	6-OHDA lessions of substantia nigra	Piroxicam (10mg/kg/day p.o) 21 day treatment	Piroxicam → ↓6-OHDA induced despair and anhedonia and normalized monoamine levels

Seo et al., 2019 (Korea)	To examine the effect of escitalopram and ibuprofen on stress-induced behaviour	Male S-D rats (12)	Adult	Chronic restraint stress (6h/day, 21 days)	Ibuprofen (40mg/kg/day i.p) 21 days	Ibuprofen → ↓stress-induced despair and normalised CORT and BDNF levels
Singh et al., 2017 (India)	To examine protective effects of losartan and ramipril on stress induced insulin resistance	Male albino Laca mice (6)	4-6 months, 20-30g	30 days Chronic restraint stress	Nimesulide (10mg/kg/day p.o), 30 days Losartan (20mg/kg/day), 30 days	Nimesulide → ↓stress induced despair Nimesulide + Losartan → ↓stress induced despair
Song et al., 2019 (China)	To examine the effect of celecoxib on neuronal injury on a rat model of depression	Male Wistar rats (18)	5-6 weeks, 160-180g	5 weeks CMS or 0.5 mg/kg/day LPS 7 days	Celecoxib (20mg/kg/day, i.p) 5 weeks treatment	Celecoxib → ↓stress-induced and LPS-induced anhedonia and despair and normalised brain PGE2 and cytokine expression Celecoxib → Ø on anhedonia and despair in control rats
Stachowicz 2020 (Poland)	To examine if indomethacin interacts with MTEP in producing antidepressant-like activity	? CD1 and C57Bl/6 mice (6-7)	?	FST or TST	Indomethacin (2 mg/kg/day i.p) for 1, 7 or 14 days Imipramine (10 mg/kg/day i.p) for 1, 7 or 14 days	Imipramine + indomethacin → ↓FST induced despair (14 but not 7 days of treatment, CD1) Imipramine + indomethacin → ↓TST induced despair (1 and 7 days of treatment, C57Bl/6) Indomethacin → Ø FST induced despair (7 and 14 days of treatment, CD1) Imipramine → Ø FST induced despair (7 and 14 days of treatment, CD1)
Warner-Schmidt et al., 2011 (USA)	To investigate a possible interaction between antidepressant agents and antiinflammatory drugs on depression related behaviour	Male C57BL/6 mice (8-16)	8-10 weeks	FST and TST	Ibuprofen (70 mg/kg/day p.o) Naproxen (140 mg/kg/day p.o) Acetylsalicylic acid (210 mg/kg/day p.o) 5-7 days treatment	The NSAIDs alone did not affect despair-like behaviour. Ibuprofen reversed antidepressant effects of citalopram, FLX, imipramine, desipramine but not tranylcipromine or bupropion in FST and TST. Naproxen and acetylsalicylic acid also reversed the antidepressant effects of citalopram. Ibuprofen, naproxen, acetylsalicylic acid and acetaminophen did not have an effect on TST or FST behaviour. Ibuprofen decreased brain IL-1β

Yamano et al., 1999 (Japan)	To investigate the mechanism of IFN-α induced depression	Male ddY mice (20)	23-36 g	IFN-α (YM643 or sumiferon)	Indomethanin (10 mg/kg, s.c), 7 days	Indomethanin → Ø TST Indomethanin → Ø in IFN-α induced despair

Table 1. Studies examining antidepressant effects of NSAIDs using rodent models

Abbreviations: Significant decrease (↓); Non-significant decrease ((↓)); Non-steroidal antiinflammatory drug (NSAID); Sprague-Dawley (S-D); Fluoxetine (FLX); Acetylsalisylic acid (ASA); No effect (Ø); Monoarthritis (MoA); Complete Freund's adjuvant (CFA); Extracellular signal-regulated kinases 1 and 2 (ERK1/2); Lipopolysaccharide (LPS); Forced swim test (FST); Wildtype (WT); Knockout (KO); Chronic mild stress (CMS);Mefenamic acid (MA); Kynurenine (KYN); Tryptophan (TRP); Not reported (?); Prostaglandin E2 (PGE2); Corticosterone (CORT); High fat diet (HFD); Interferon (IFN); Interleukin (IL); Amyloyd beta (Aβ); Intracerebroventricular (ICV); Secreted phosphoprotein 1 (Spp1); Cyclophosphamide (CP); Ovariectomized (OVX); Cyclooxogenase (COX); Bacillus Calmette-Guerin (BCG); Tail suspension test (TST); 6-Hydroxydopamine (6-OHDA); Brain-derived neurotrophic factor (BDNF); Intraperitoneal (i.p); Intragastric